

Cloud Journey Notes (Days 1–5) – Detailed Version

Day 1 – Linux Basics

Operating System (OS): Software that acts as a bridge between hardware and applications. It manages memory, storage, processes, and devices.

Linux: A popular open-source operating system widely used in servers, cloud computing, DevOps, and cybersecurity.

Kernel: The core component of Linux that communicates directly with hardware and allocates resources.

Shell: A command interpreter that accepts user commands and passes them to the operating system.

GUI vs CLI: GUI uses windows and icons, while CLI uses text commands. Cloud engineers prefer CLI because it is faster, scriptable, and works efficiently on remote servers.

Important Commands: pwd, ls, cd, mkdir, rmdir, touch, cat, clear.

Day 2 – Permissions & Users

Linux Security Model: Linux controls access using permissions and ownership.

Permission Types: Read (r) – View file contents. Write (w) – Modify file contents. Execute (x) – Run a file or enter a directory. **Permission Groups:** Owner – Creator of the file. Group – Users belonging to the same group. Others – Everyone else. **Useful Commands:** ls -l, chmod, chown.

Important Concept: Directories require execute permission to enter them. Even if a directory can be read, it cannot be accessed without execute permission.

Day 3 – Processes & Services

Process: A running instance of a program. Every process receives a unique Process ID (PID).

Foreground Process: Uses the current terminal and blocks further commands until it finishes.

Background Process: Runs independently and allows continued terminal usage.

Service (Daemon): A special background process that provides system functionality continuously, such as SSH or cron.

Monitoring Commands: ps, ps aux, top, htop.

Process Management: kill sends a signal to stop a process, while kill -9 forcefully terminates it.

Key Idea: Monitoring CPU, memory, and running services is essential for maintaining reliable systems.

Day 4 – Logs & Troubleshooting

Logs: Records generated by the operating system and applications to capture events, warnings, and errors.

Why Logs Matter: Logs help engineers diagnose issues based on evidence rather than assumptions.

Important Commands: journalctl, journalctl -f, systemctl status, dmesg.

Troubleshooting Workflow: Identify the issue. Check service status. Inspect logs. Determine root cause. Apply fix and verify. **Key Idea:** Reading logs is one of the most valuable skills in Linux, Cloud, and DevOps.

Day 5 – Package Management

Package: A bundle containing software, configuration files, and dependencies.

Package Manager: A tool used to install, update, search, and remove software efficiently.

APT Commands: apt update – Refreshes package information from repositories. apt upgrade – Installs available updates. apt install – Installs software packages. apt remove – Removes installed packages. apt search – Searches available packages. **Repositories:** Online locations that store trusted software packages.

Benefits: Package managers automatically handle software dependencies, making installation safer and easier.